**Worksheet 9.2**

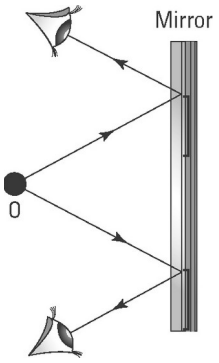
*Science Quest 9:* pages 314–19

# Reflection and scattering of light

Student: ..... Class: .....

## Locating the virtual image

**1.** An object (O) is observed by looking at the rays of light reflecting off a plane mirror. The mirror consists of a plate of glass with a silver back surface mounted on a sheet of wood.




(a) Use a ruler and pencil to extend each reflected ray (using dotted lines) until they intersect behind the mirror. Label the point of intersection as 'virtual image'.

(b) Complete the sentence:

The object in front of a reflecting surface and a virtual image are the …… distance from a mirror.

(c) Draw the *normal* to the mirror surface and measure the incident and reflected angles for the ray above the object, O.

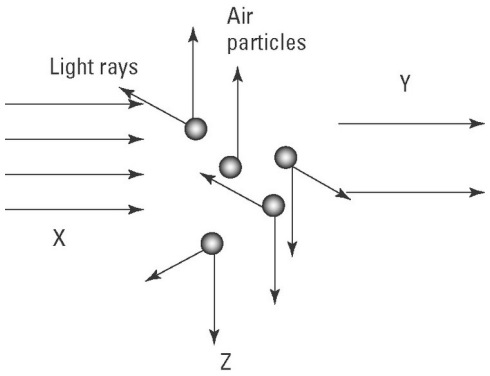
## Light and air

**2.** The diagram below shows the collision of light rays with particles of air in the atmosphere. Use the letters X, Y and Z to identify the following rays.

Scattered: .....

Incident: .....

Transmitted: .....




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